Digital Bluetooth Stethoscope

CA1

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OVERVIEW

1 Introduction&Literature Review

2 Hardware Design

3 Software Implementation

4 Edge AI Design

Introduction

&Literature Review

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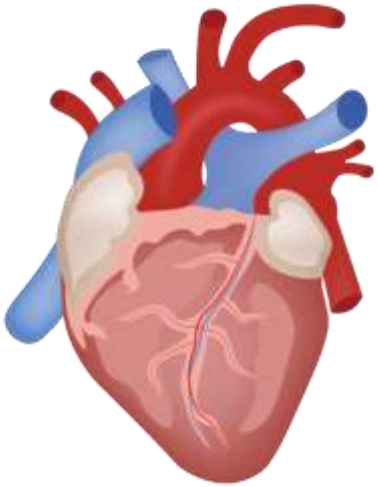
Introduction

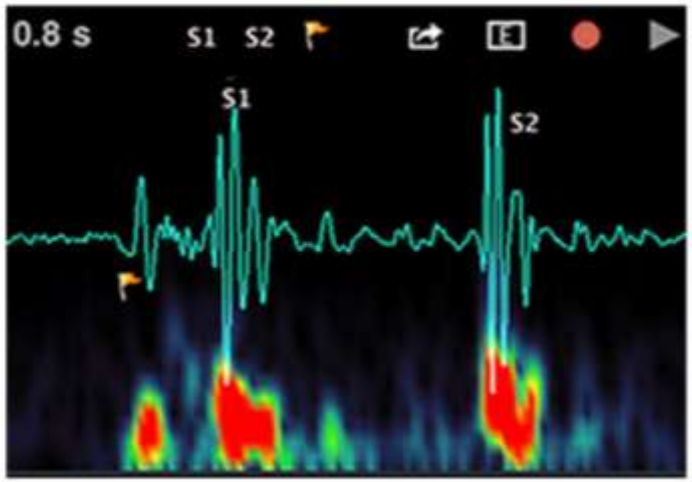


Cardiovascular diseases (CVDs) are the leading cause of death globally.

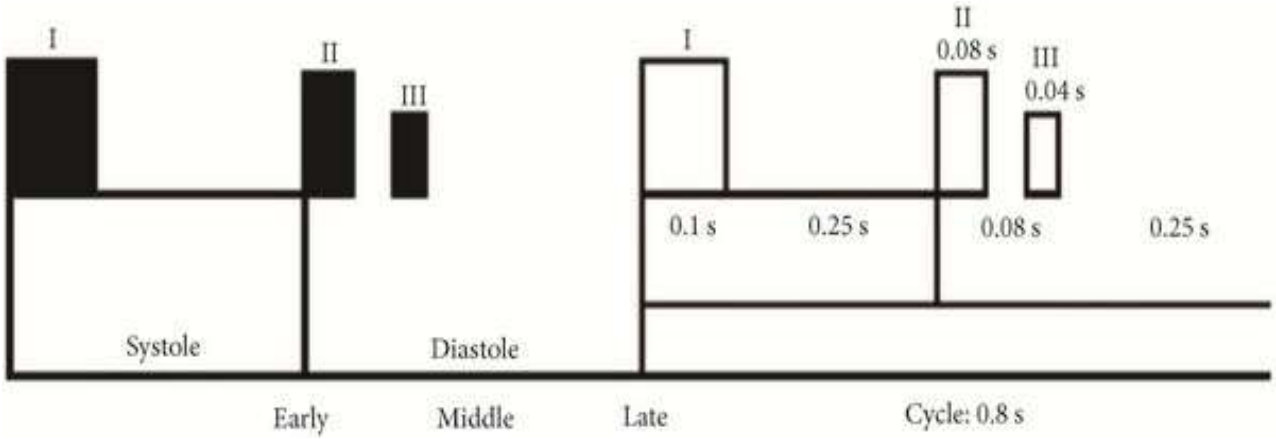
- About **30%** of people died from CVDs
- Over **3/4** of CVD deaths in low- and middle-income countries.

→ Lack of primary health care programs

Introduction



1. **Heart Sounds: S1， S2**
2. **Murmurs**
3. **Additional Heart Sounds： S3， S4**



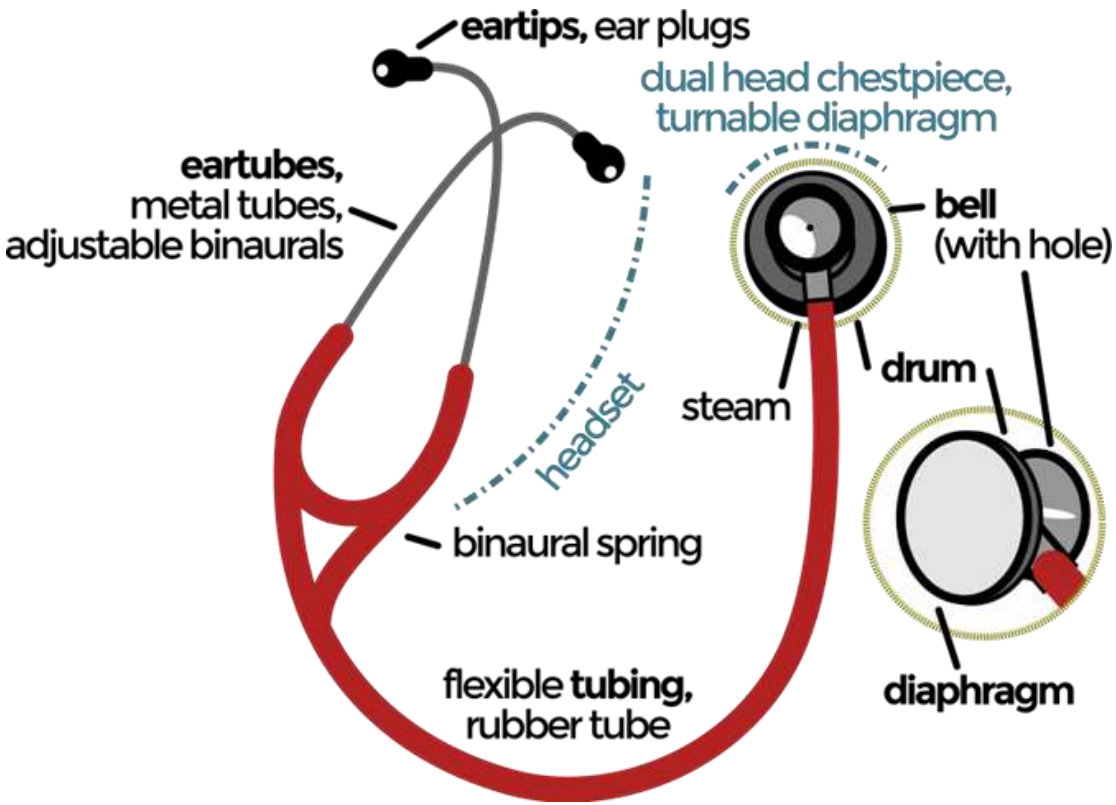
Systole	0.35 s
Diastole	0.45 s
A complete cycle	0.8 s

Frequency:

S1, S2 : 20 – 400 Hz

Murmurs : 600 Hz at most

Introduction



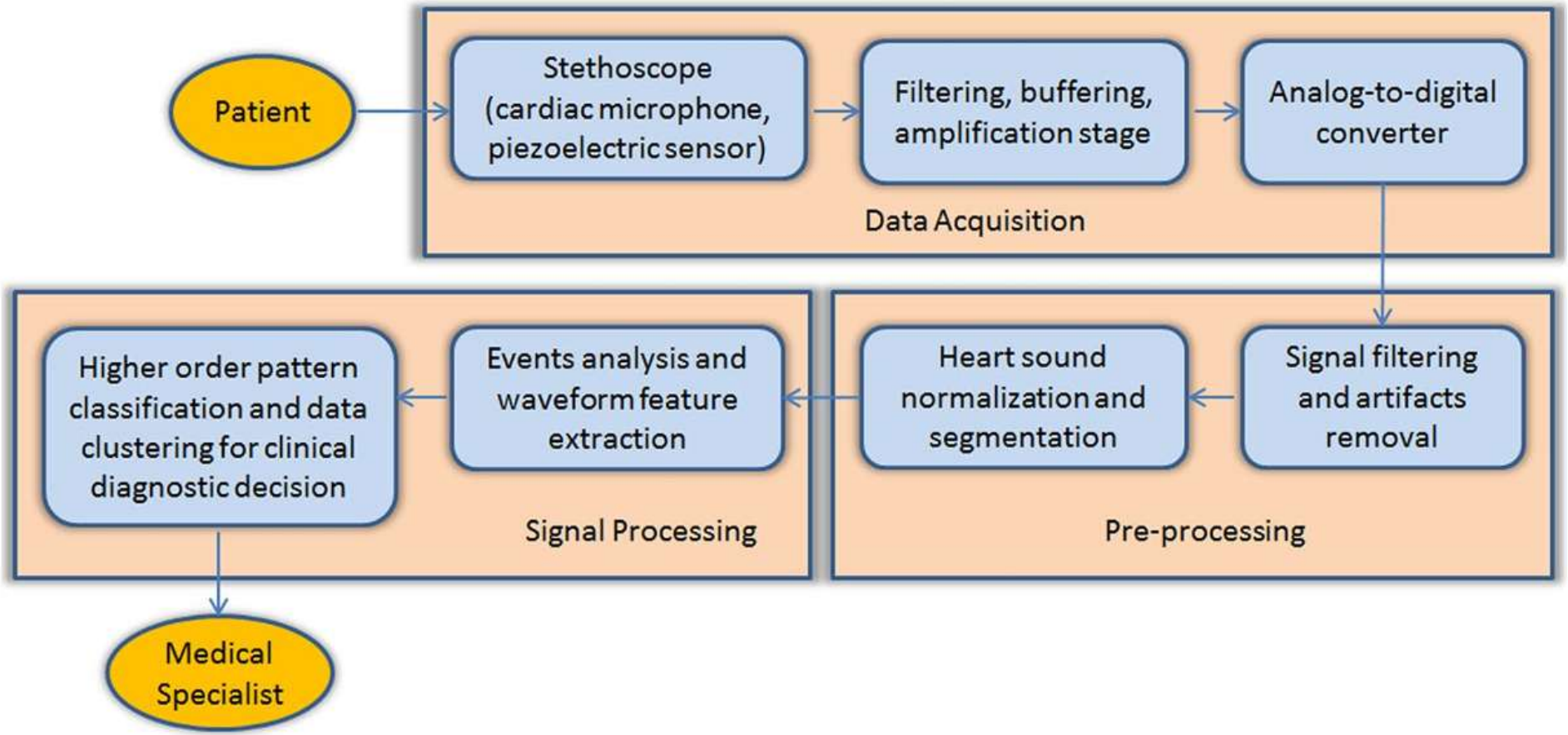
Non-invasive
low-cost
portable

Three types:

Analogue, Electronic, Digital

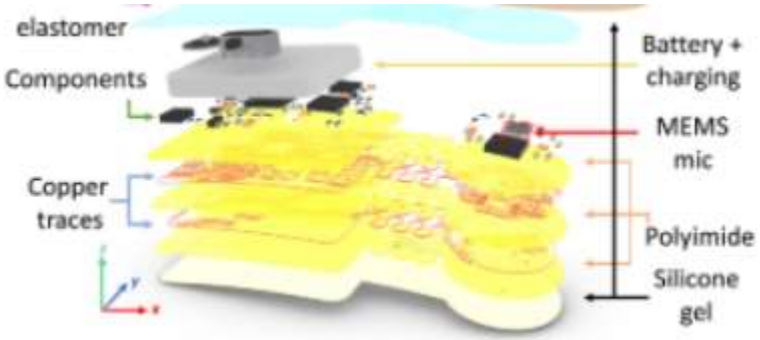
Literature Review

[1] Leng S, et al. The electronic stethoscope[J]. BioMedical Engineering OnLine, 2015, 14: 66.



Literature Review

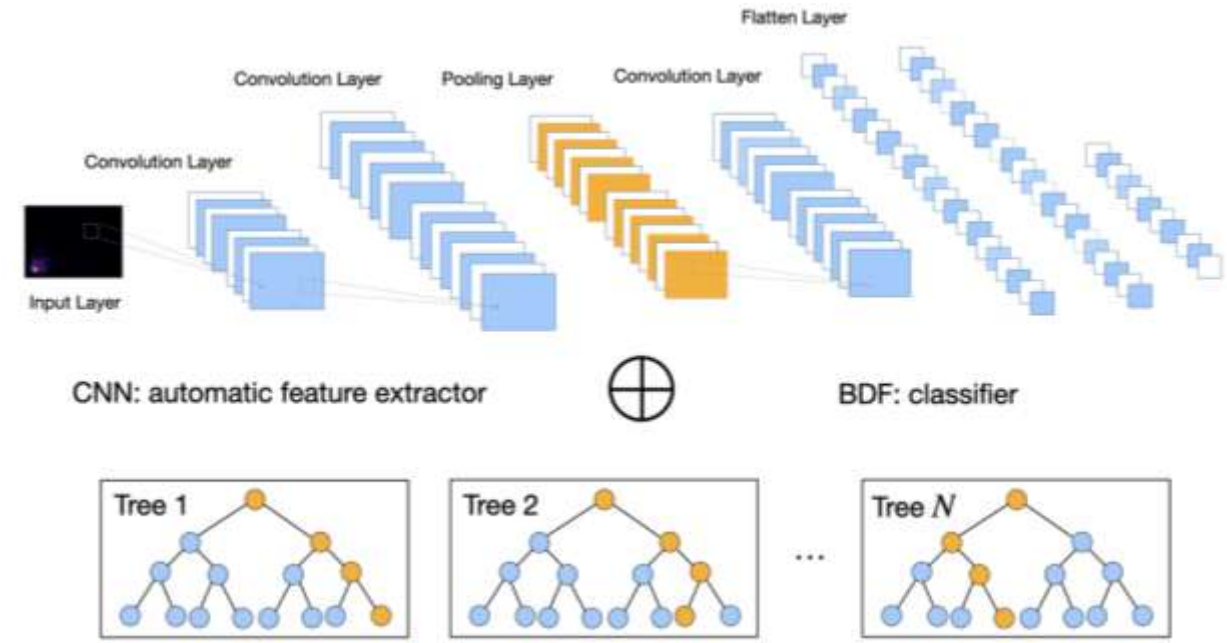
[2] Lee S H, et al. Fully portable continuous real-time auscultation with a soft wearable stethoscope[J]. Science Advances, 2023.



- a soft wearable stethoscope
- Combined with machine learning

Literature Review

[3] Zhang M, et al. A Low-Cost AI-Empowered Stethoscope and a Lightweight Model for Detecting Cardiac and Respiratory Diseases[J]. Sensors, 2023, 23(5): 2591.



- A digital stethoscope built on Raspberry Pi Zero 2W and a MEMS mic costs 25 USD

- CNN-RF model

Market Product (State-Of-The-Art)

Eko Core 500

Drawbacks:

- 200 US dollars
- 186 grams
- Designed for doctor
- Regulatory and usage restrictions

Scheme iteration and hardware design

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Overall system framework

transmitting terminal

Internal processing flow of MCU

- Drive Read
- The decimation algorithm
- Digital gain amplification
- LPF（if）
- BLE protocol stack\
  nGATT Server,Notify

MCU

Digital audio stream

MEMS Digital microphone

sound wave

Cardiopulmonary sound signal

lithium battery +LDO

3.3V supply

PCM

Raspberry Pi

bluetooth reception

- Scan link subscribe and receive
- PCM reconstruction

preprocessing

- denoise（DWT）
- band-pass filtering
- resample
- normalization

AI reasoning

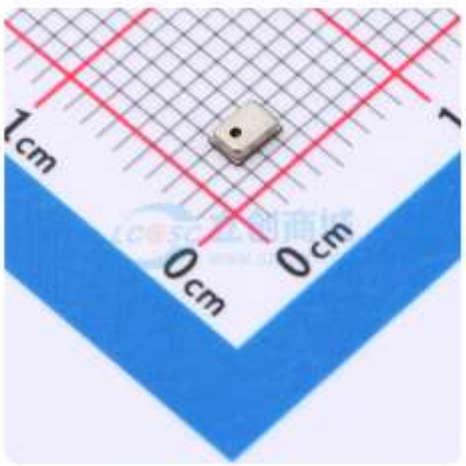
- data pre-processing
- feature extraction
- Load the mode
- Result display/upload

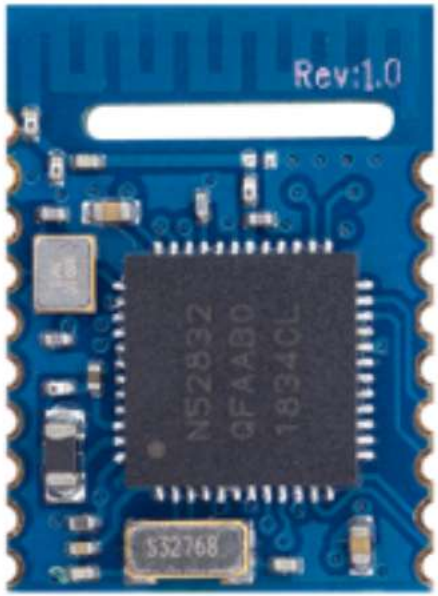
model training

if

- Complex reasoning
- Cloud-based consultation
- push notification

The initial plan： LMD2718T+nRF52810





Maintain acoustic properties

Suitable for medical use

small volume

low cost

low power consumption

pain spot:

Only supported UART

The configuration of the PDM driver is extremely complicated.

The development cycle is uncontrollable.

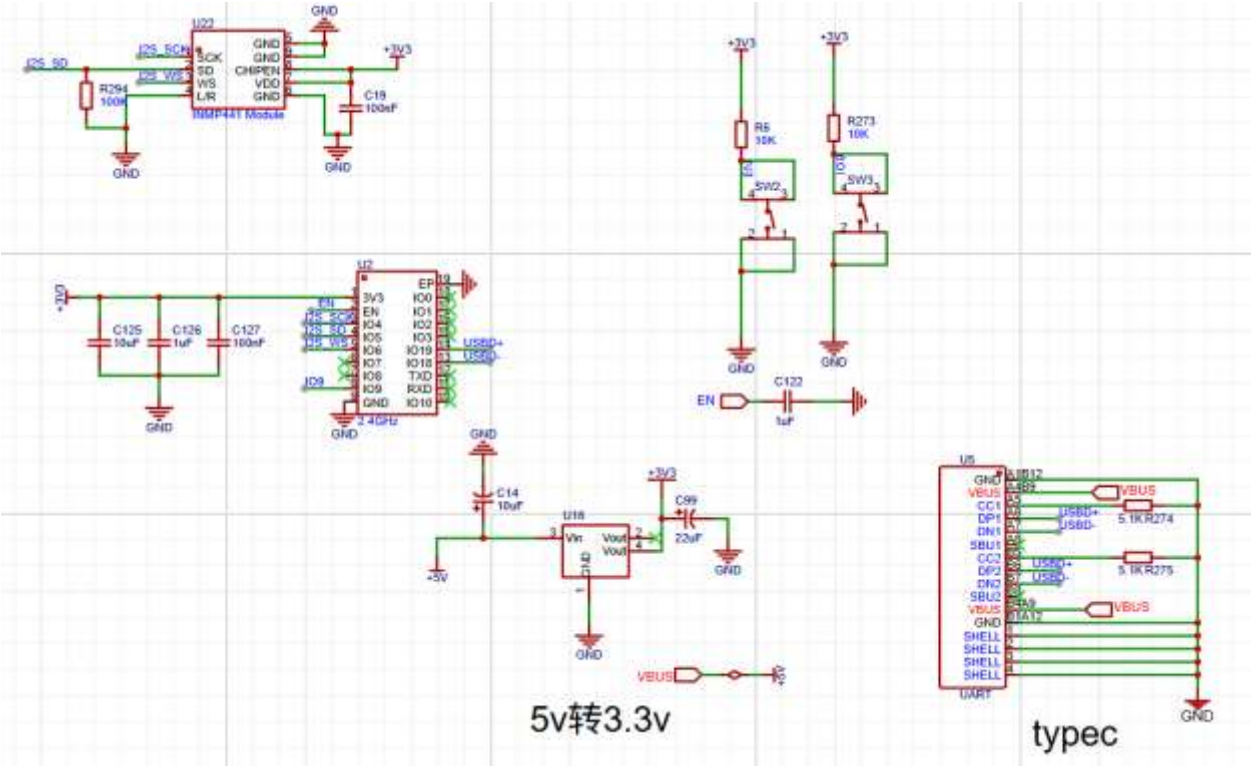
improvement project:INMP441 + ESP32

Parameter	LMD2718T	INMP441
output interface	PDM	I2S
SNR	54 dBA	61 dBA
sensitivity	-27 dBFS	-26 dBFS
current drain	0.75 mA	1.4 mA
Data processing difficulty	High(PDM→PCM)	Low

ESP32-C3-WROOM-02

	nRF52810	ESP32-C3
Main frequency speed	64Mhz	160Mhz
SRAM	24KB	400KB
Flash	192KB	4MB
audio interface	UART	I2S
Peripheral resources	resource-constrained	Rich (22 GPIOs, SPI, UART, I2C, etc.)

Hardware PCB implementation



Software and algorithm implementation

- Audio Sampling
- DSP signal processing
- BLE Bluetooth data broadcasting

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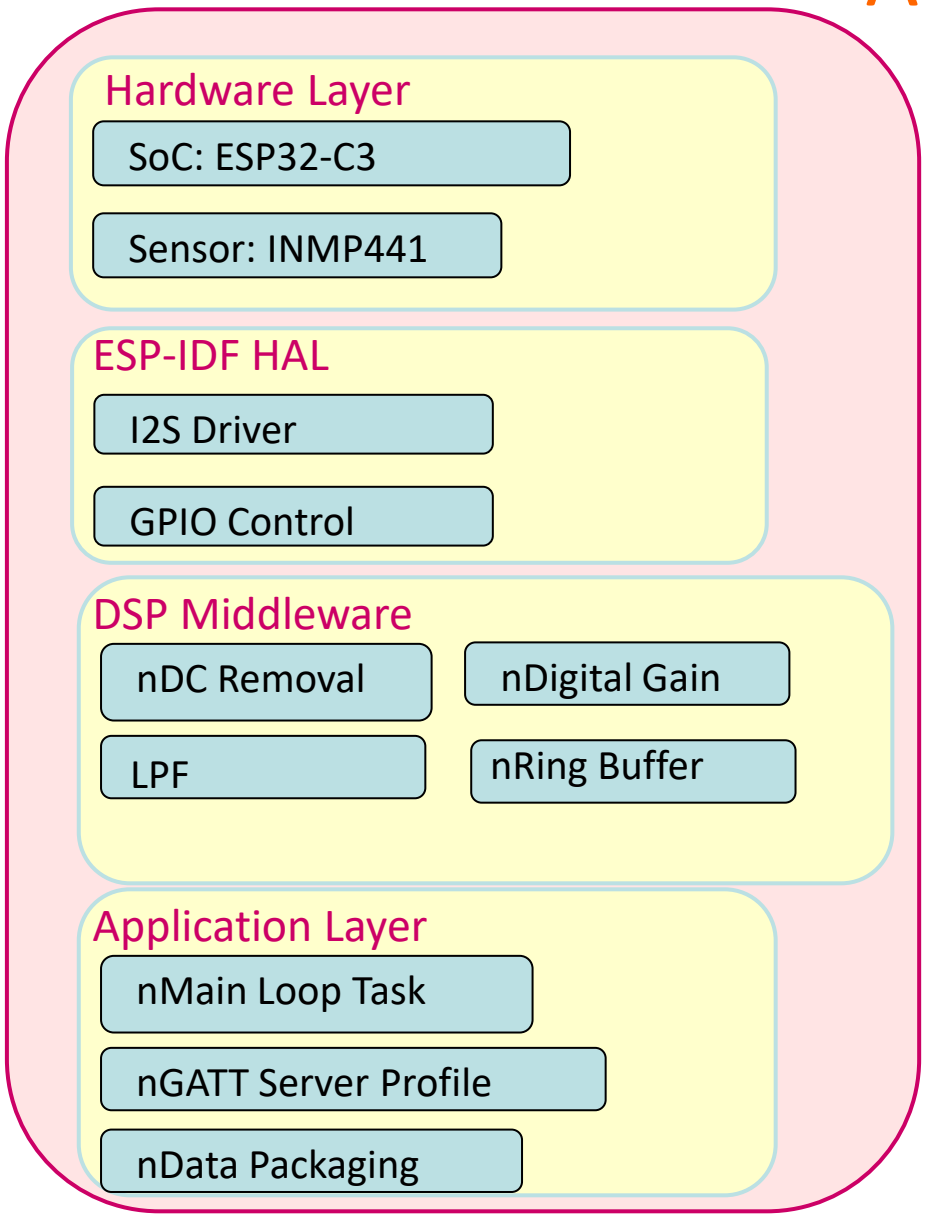
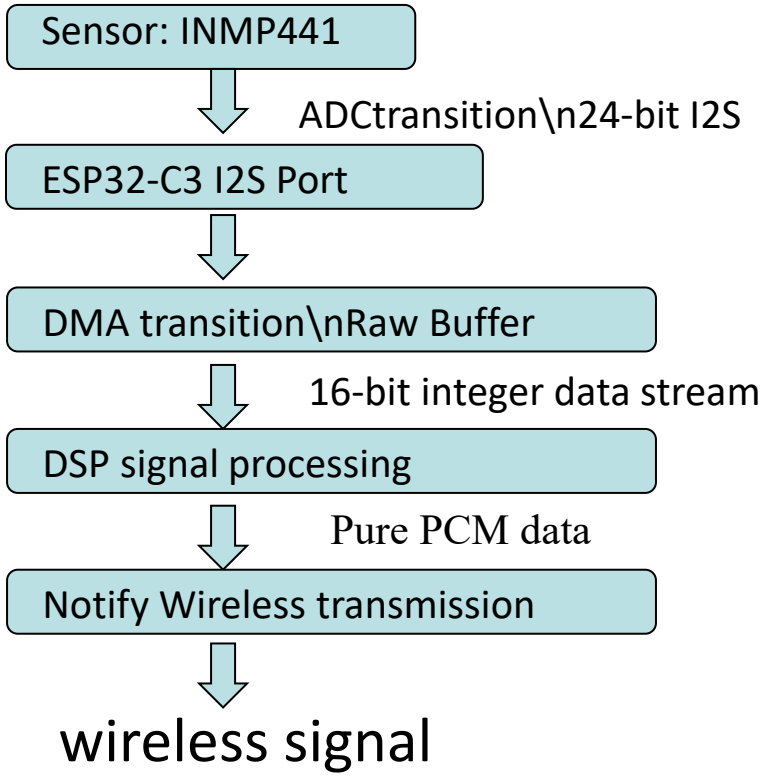
Software Architecture

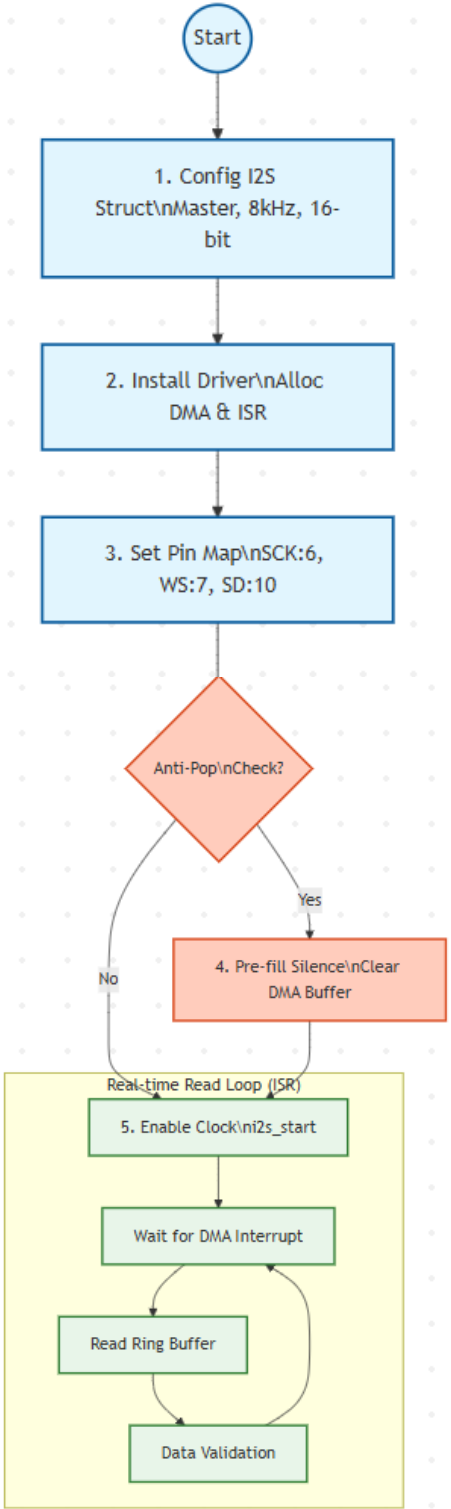
Software Layered Architecturel



Audio Data Flow Pipeline



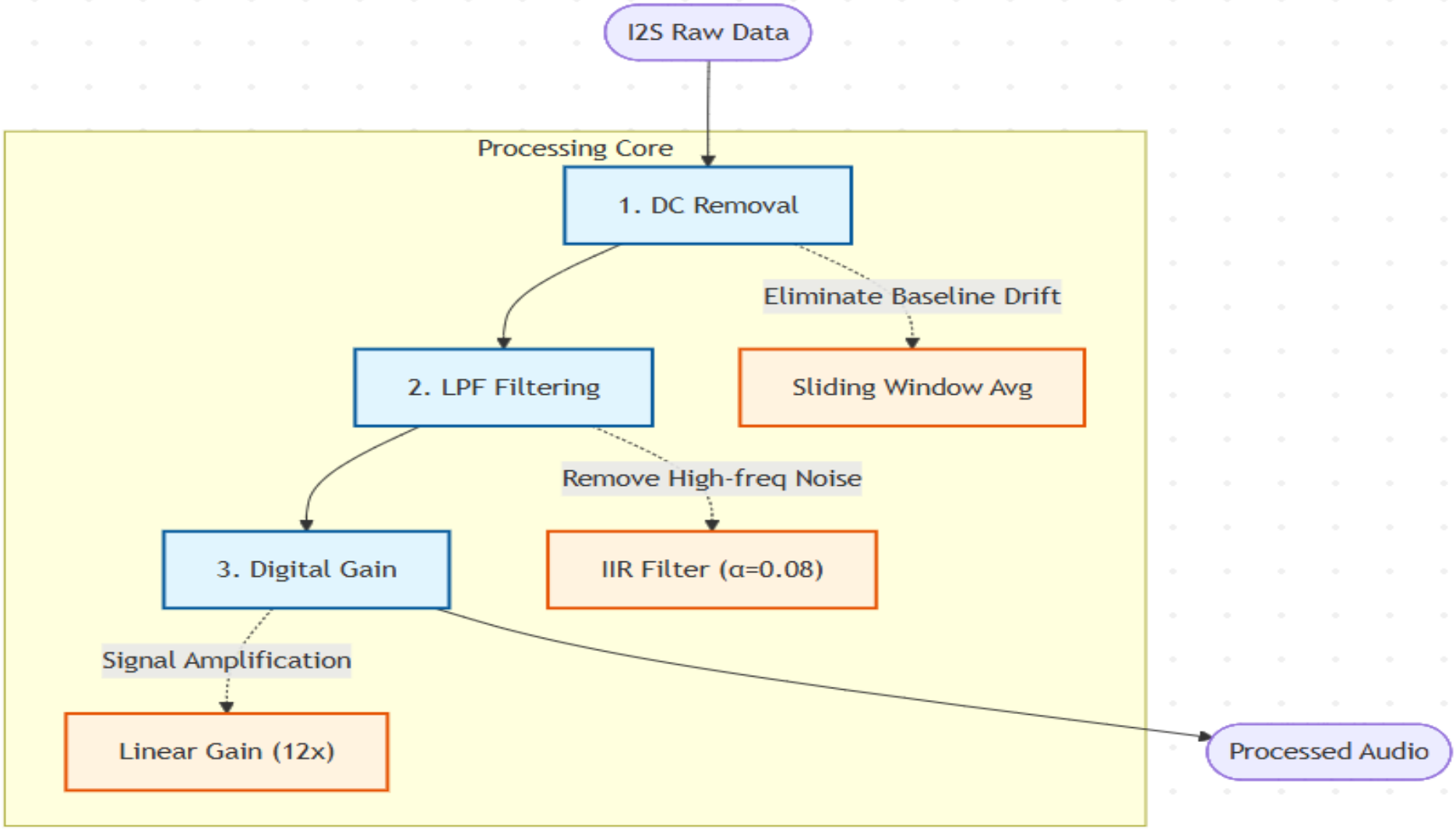
Developed based on the Arduino framework



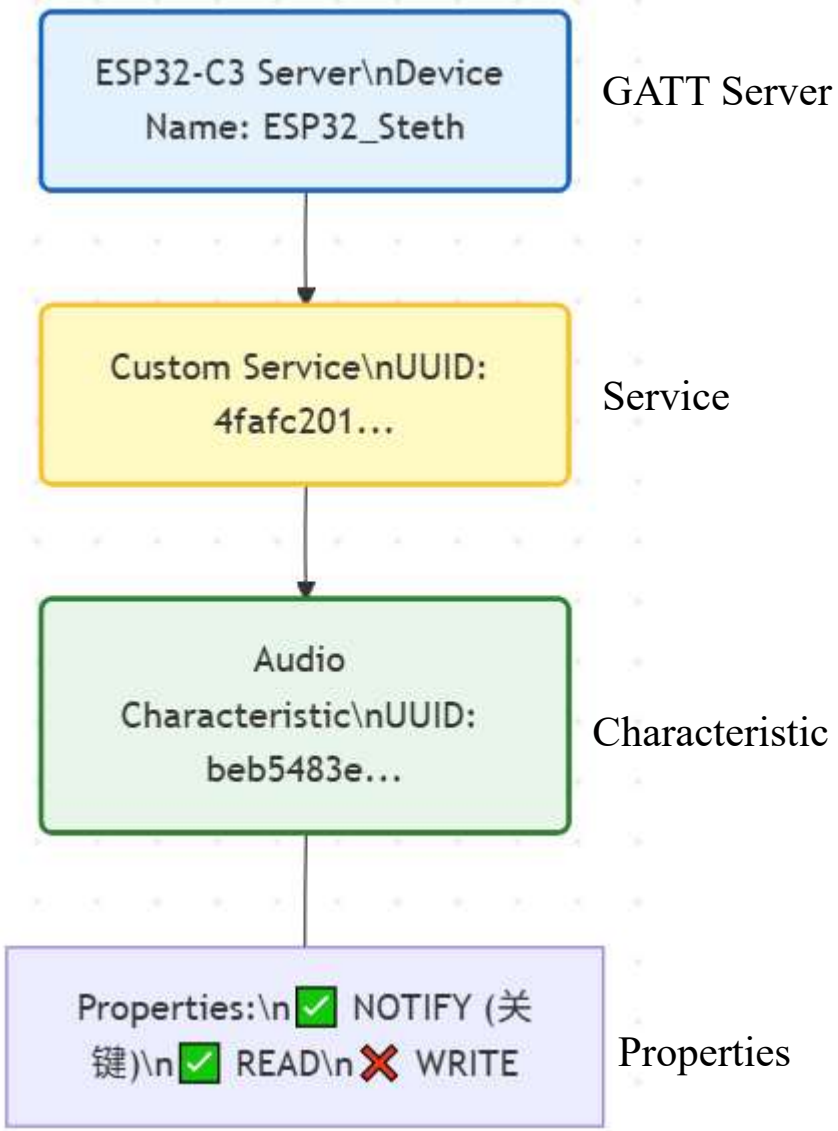
I2S Audio Driver Mechanism

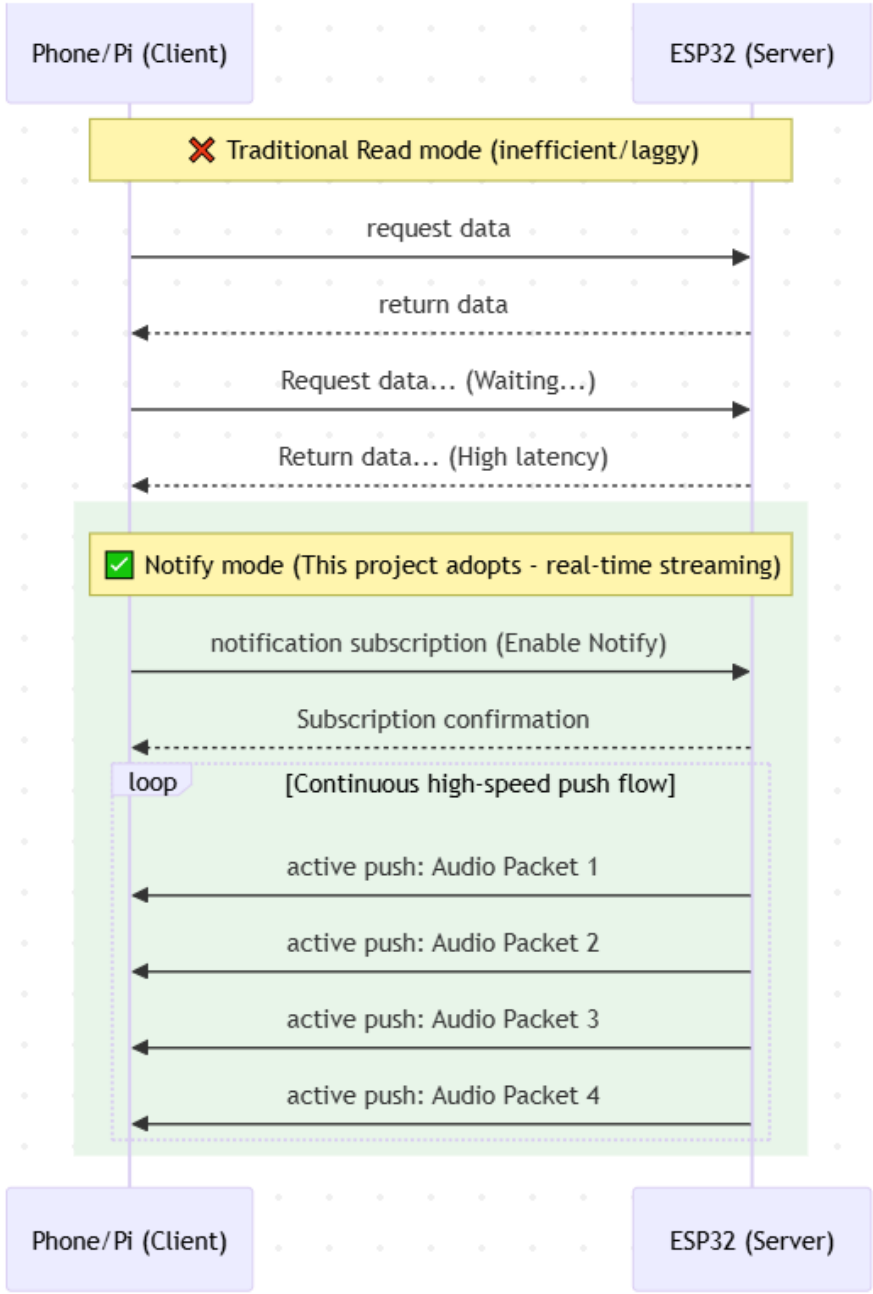
Parameter	value	Design Rationale
communication mode	Standard I2S Master	Generate clock signals to ensure stable timing
sampling frequency	8000 Hz	Meet the requirements for heart sound (20-500Hz) collection and reduce the Bluetooth load
DMA Buffer	8 x 128 Bytes	Multi-level buffering prevents data overwriting and ensures audio continuity
data format	16-bit PCM	Adapt to BLE MTU transmission units to optimize bandwidth utilization

Digital Signal Processing



BLE Communication





Notify's streaming mechanism

Current achievements

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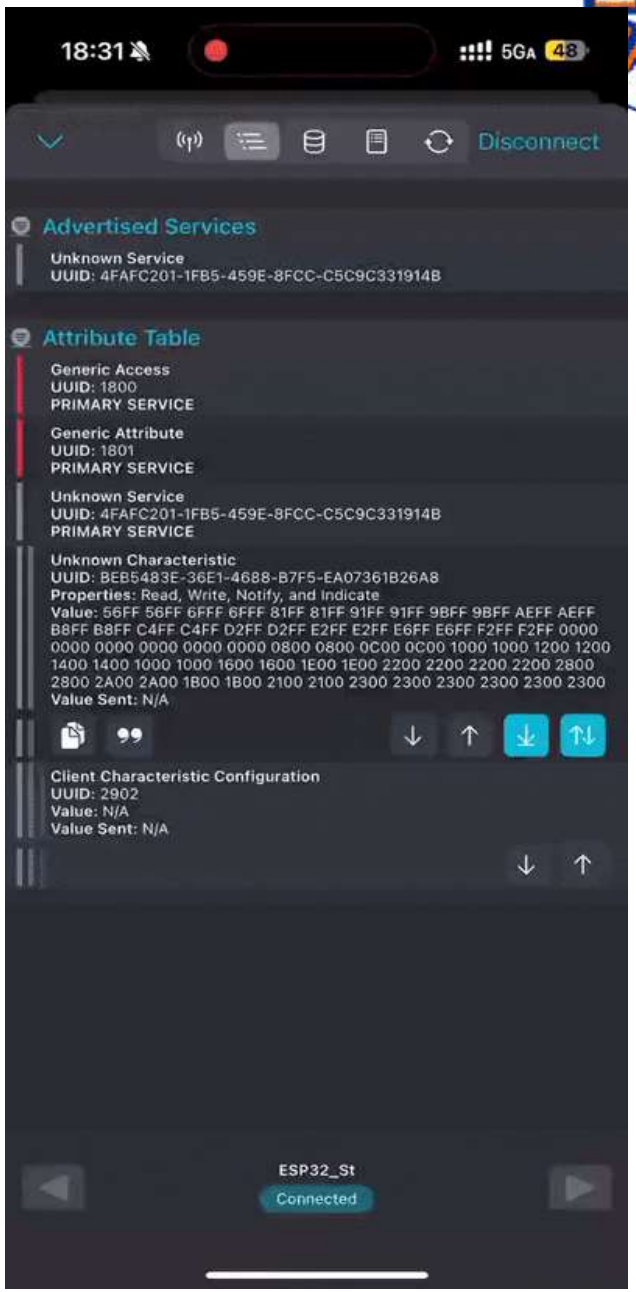
COM7
value 1
Interpolate
STOP
800
600
400
200
0
-200
-400
-600
7461
7473
7485
7497
7510
Type Message
SEND
New Line
9600 baud


18:31
5G
48
Disconnect
Advertised Services
Unknown Service
UUID: 4FAFC201-1FB5-459E-8FCC-C5C9C331914B
Attribute Table
Generic Access
UUID: 1800
PRIMARY SERVICE
Generic Attribute
UUID: 1801
PRIMARY SERVICE
Unknown Service
UUID: 4FAFC201-1FB5-459E-8FCC-C5C9C331914B
PRIMARY SERVICE
Unknown Characteristic
UUID: BEB5483E-36E1-4688-B7F5-EA07361B26A8
Properties: Read, Write, Notify, and Indicate
Value: 56FF 56FF 6FFF 6FFF 81FF 81FF 91FF 91FF 9BFF 9BFF AEFF AEFF B8FF B8FF C4FF C4FF D2FF D2FF E2FF E2FF E6FF E6FF F2FF F2FF 0000 0000 0000 0000 0000 0000 0800 0800 0C00 0C00 1000 1000 1200 1200 1400 1400 1000 1000 1600 1600 1E00 1E00 2200 2200 2200 2200 2800 2800 2A00 2A00 1B00 1B00 2100 2100 2300 2300 2300 2300 2300 2300
Value Sent: N/A
Client Characteristic Configuration
UUID: 2902
Value: N/A
Value Sent: N/A
ESP32_St
Connected

Design and implementation of Edge AI

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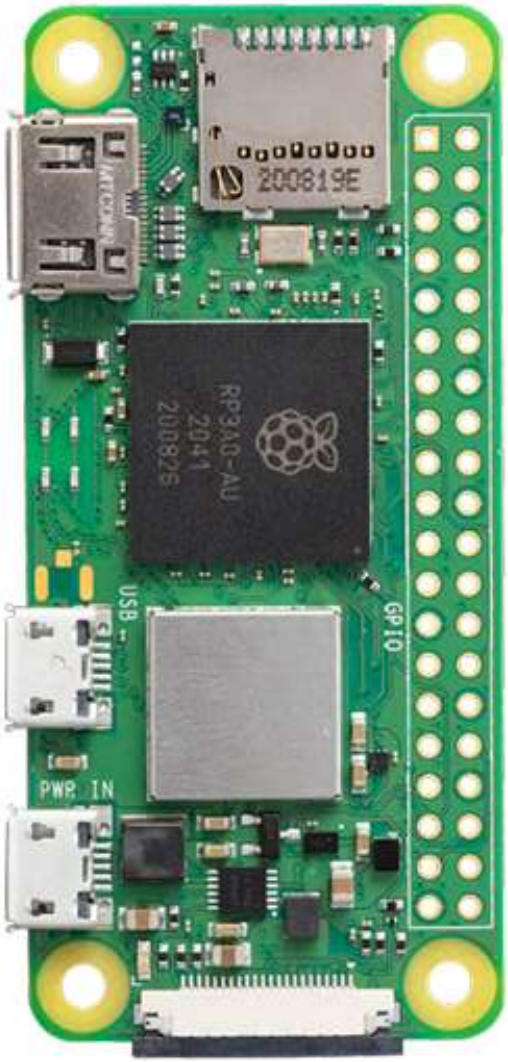


Form factor:	65mm × 30mm
Processor:	Broadcom BCM2710A1, quad-core 64-bit SoC (Arm Cortex-A53 @ 1GHz)
Memory:	512MB LPDDR2
Connectivity:	• 2.4GHz IEEE 802.11b/g/n wireless LAN, Bluetooth 4.2, BLE, onboard antenna • 1 × USB 2.0 interface with OTG • HAT-compatible 40-pin I/O header footprint • microSD card slot • Mini HDMI port • CSI-2 camera connector
Video:	• HDMI interface • Composite video
Multimedia:	• H.264, MPEG-4 decode (1080p30) • H.264 encode (1080p30) • OpenGL ES 1.1, 2.0 graphics
Input power:	5V DC 2.5A
Operating temperature:	-20°C to +70°C

Setup of Raspberry Pi

- System Image Flashing

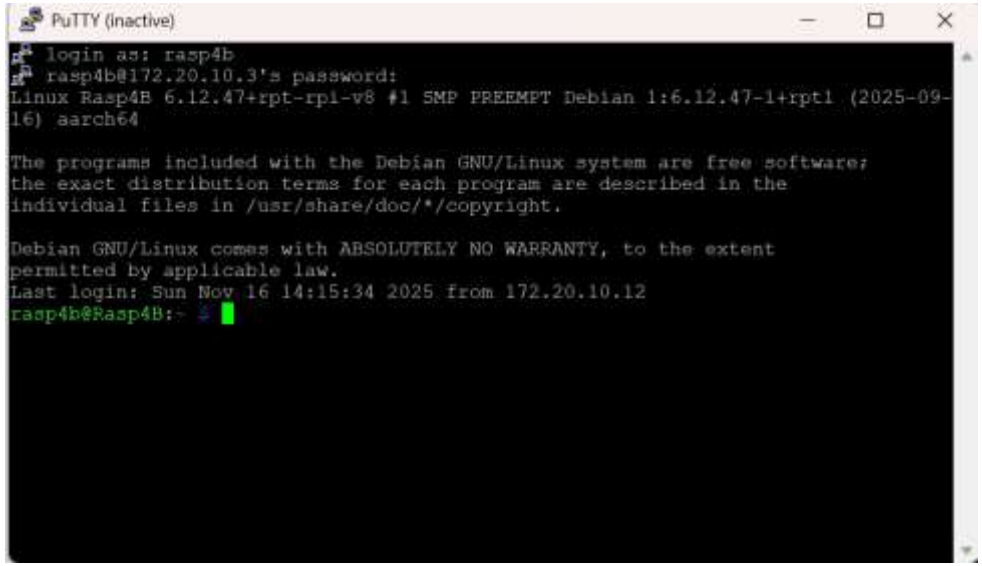
- Setup of SSH & WIFI

ssh.txt

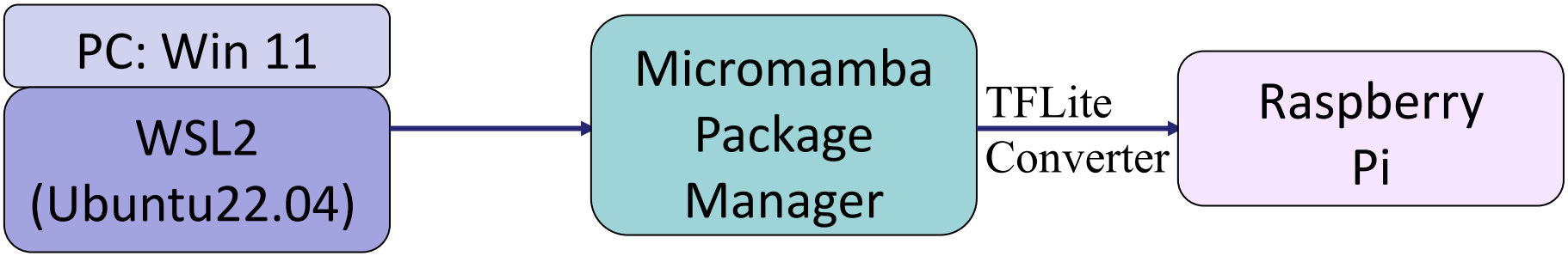
wpa_supplicant.conf

→ microSD

- SSH connection



Setup of Environment



Dataset

- PASCAL Classifying Heart Sounds Challenge 2011 (CHSC2011)
  set_a, set_b
  Classes: five (artifact, extrahls, murmur, extrastole, normal)
  Time length: 1 ~ 9 seconds
  Sampling rate: mostly 4k Hz

- The PhysioNet/Computing in Cardiology Challenge 2016
  3,153 samples
  Classes: two (normal, abnormal)
  Time length: 5 ~ 120 seconds
  Sampling rate: 2k Hz

Ablation

For feature:

	Mel	Bispectrum	Wavelet
Nature	Speech-like Time-Frequency Representation	Multi-scale Time-Frequency Analysis	Higher-Order Spectra
Info	Frequency Band Energy + Temporal Structure	Non-stationary Signals & Transient Structures	Phase Coupling & Non-Gaussianity
Best Acc	**0.8370**	0.8238	0.6432
Convergence	**Fast**	Slightly Slow	Slow
Stability	Minor fluctuations in later epochs	Very Stable	Unstable

Ablation

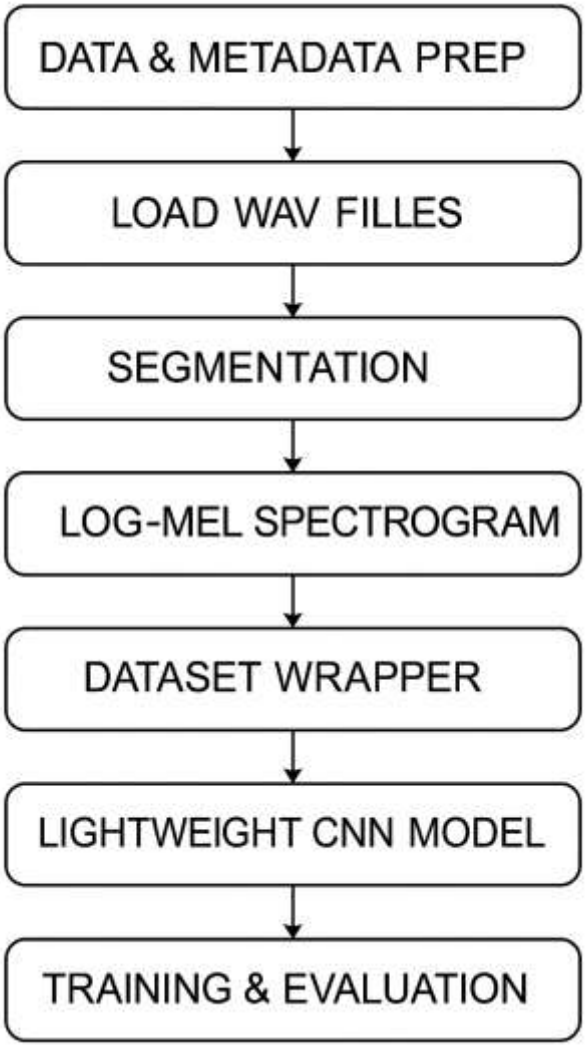
For model:

	Lightweight CNN	Audio Transformer
Machanism	Local Conv	Attention
Data View	Image-based	Sequence-based
Parameters	Small (~15k)	Large (~100k)
Feature Extraction	Hierarchical	Global Interaction
Deployment Feasibility	Excellent	Moderate
Loss	**0.5897**	0.9019
Accuracy	**0.7302**	0.6562

Code

https://github.com/RaeXuu/FYP

Pipeline:



Structure:

Pre-process

- load_metadata.py
- load_wav.py
- filter.py
- segment.py
- mel.py

Model

- lightweight _cnn.py

Train

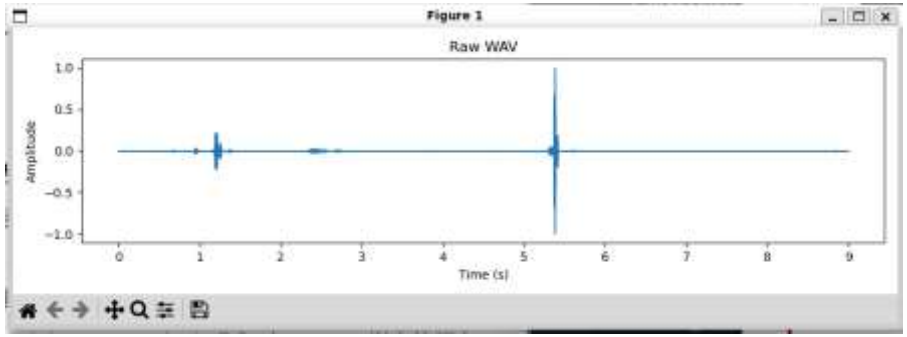
- dataset_mel.py
- train_lightweight.py
- train_lightweight_wi th_test.py

Inference

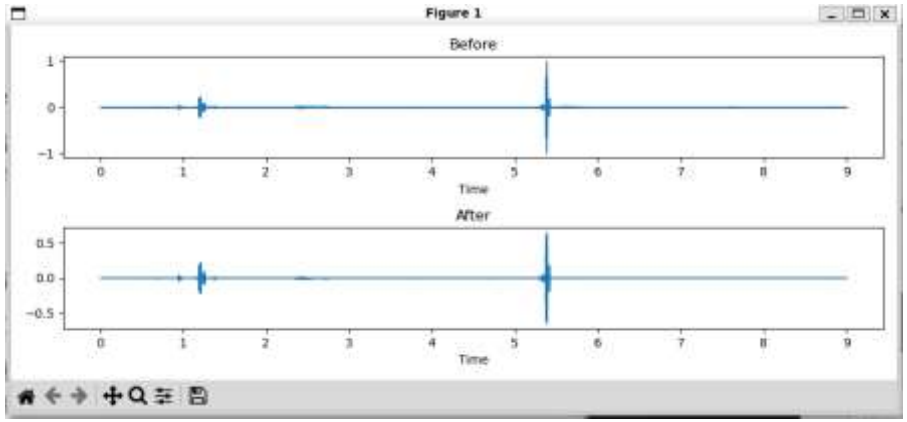
- benchmark_ model.py
- run_inference .py

Pre-process Visualize

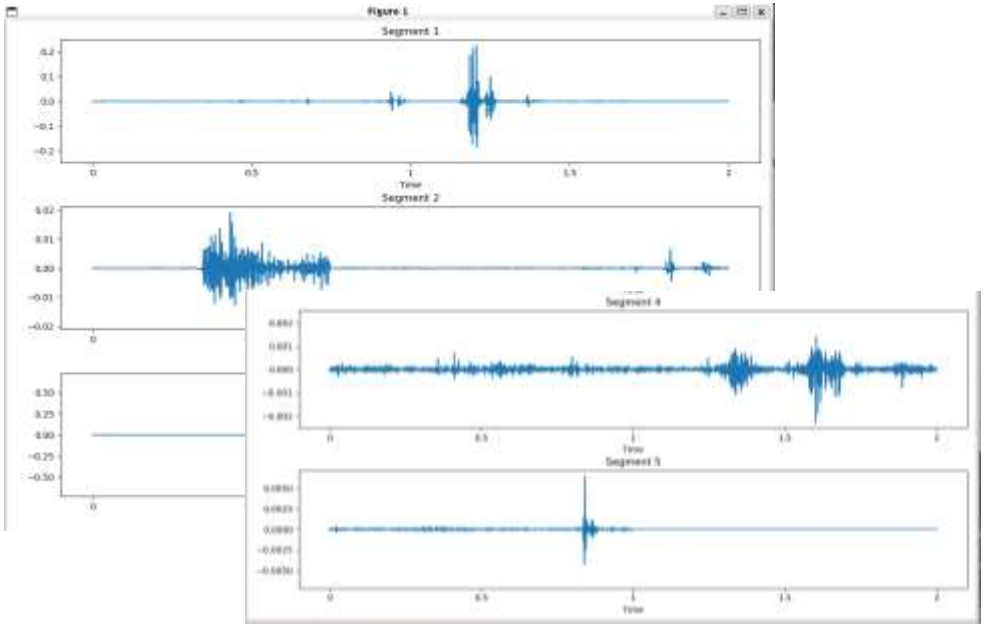
- **load_wav.py**



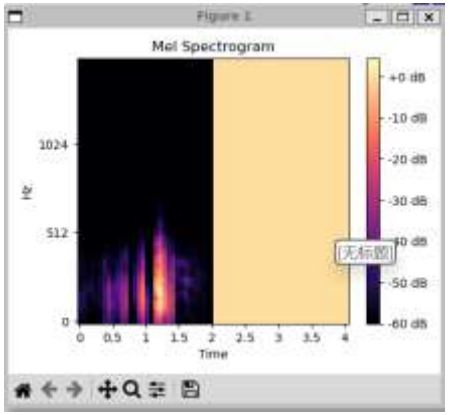
- **filter.py**



- **segment.py**



- **mel.py**

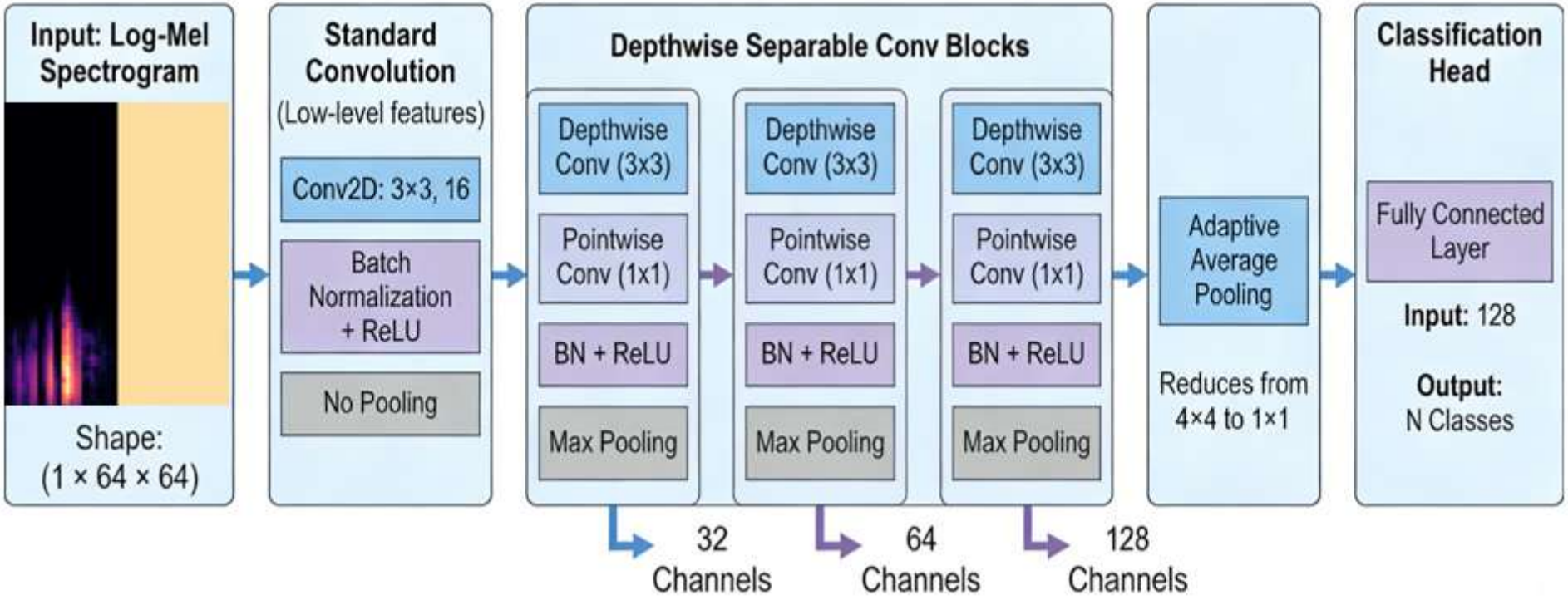


Model & Train

- **dataset_mel.py**

  —> PyTorch Dataset

- **lightweight_cnn.py**



Train

- **train_lightweight.py**

optimizer: Adam
batch size: 16
epoch: 25

- **train_lightweight_with_test.py**

train/validation/test split: **8:1:1**

One Epoch Evalutation:
Train Loss, Acc
Validation Loss, Acc
Confusion Matrix
Classification Report
(precision, recall, f1-score, support)

Result:

```
=========================================
[TEST] Loading best model and evaluatin
=========================================

Test  Loss: 0.5751 | Test  Acc: 0.8798

[Test] Confusion Matrix:
[[ 16   0   0   0   0]
 [  1  19   0   0   3]
 [  0   0  20   0   4]
 [  0   0   0  72  17]
 [  0   0   2  15 183]]
```

Parameter Tuning

➡ numerous ablation experiment

Parameter	Value
Sample Rate	2000 Hz
Segment Length	2.0 s
Overlap	50%
Bandpass Low	20 Hz
Bandpass High	400 Hz

Parameter	Value
n_fft	256
win_length	256
hop_length	96
n_mels	32
fmin	20 Hz
fmax	400 Hz
power	2.0

Inference & Benchmark

- **run_inference.py**

Logits are converted to a probavility via **Softmax**
the final classification result is obtained using **Argmax**.

```
(FypProj) rae@Rae:/mnt/d/FypPr
Predicted class: 0
Probabilities: [9.9980062e-01
```

- **banchmark_model.py**

```
(FypProj) rae@Rae:/mnt/d/FypProj$ pyt
==== Deployment Benchmark ====
[Params] 13,253
[File ] 0.07 MB
[MACs ] 35.19 M
[CPU  ] Latency = 0.962 ms / sample
[TPS  ] 1039.02 samples/s
================================
```

Category	Metric
Model Size	Nums of Parameters
	Weight File Size (MB)
Latency	(ms)
Complexity	MACs / FLOPs
Energy Consumption	Time per Inference
Accuracy	FP32 / INT8
Runtime Memory	Activations memory